WORD-COMPATIBLE GRAPHS AND GROVE POLYNOMIALS

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ABSTRACT.

1. WC GRAPHS

Definition 1.1. A *WC graph* is simply a finite subset $W \subseteq \mathbb{N} \times \mathbb{N}$ with no additional restrictions. We impose a total ordering on these pairs by declaring that $(a, b) \leq (c, d)$ if $a < c$ or $a = c$ and $b \geq d$. We define the *weight* of a WC graph W to be the sequence

$$\mathtt{wt}(W) = (i_1, \ldots, i_n)$$

where i_j is the number of elements of W in row j. We define the *permutation* of a WC graph W to be the permutation

$$\mathtt{perm}(W) = s(w_1) \bullet \cdots \bullet s(w_n)$$

where

$$w_i = (a_i, b_i)$$

is the ith element in the imposed ordering, and

$$s(a_i, b_i) = s_{a_i+b_i-1}$$

with $\bullet$ being the Hecke product. The *word* of W is

$$(s(w_1), \ldots, s(w_n))$$

Define $\mathcal{WC}(w)$ to be the set of all WC graphs W such that $\mathtt{perm}(W) = w$.

The following is well-known.

Theorem 1.2. *We have*

$$\mathfrak{G}_w^{(\beta)}(x) = \sum_{W \in \mathcal{WC}(w)} \beta^{|W| - \ell(w)} x^{\mathtt{wt}(W)}$$

2. GROVE POLYNOMIALS AND WC GRAPHS

Definition 2.1. A labeled indexed forest (F, P) is an indexed forest F with a function (labeling) $P : \mathrm{IN}(F) \to \mathbb{N}$ satisfying the following properties:

(1) $P(\mathrm{left}(u)) < P(u)$ for all $u \in \mathrm{IN}(F)$.
(2) $P(u) \leq P(\mathrm{right}(u))$ for all $u \in \mathrm{IN}(F)$.
(3) $P(u) \geq \rho_F(u)$ for all $u \in \mathrm{IN}(F)$.
(4) $P(u) \leq \lambda_F(u)$ for all $u \in \mathrm{IN}(F)$.

The set $\mathrm{Komp}_P(F)$ is the set of functions $\kappa : \mathrm{IN}(F) \to 2^{\mathbb{N}}$ such that

(1) $\max(\kappa(u)) \leq P(u)$ for all $u \in \mathrm{IN}(F)$.
(2) $\max(\kappa(u)) \leq \min(\kappa(\mathrm{left}(u)))$ for all $u \in \mathrm{IN}(F)$.
(3) $\max(\kappa(u)) < \min(\kappa(\mathrm{right}(u)))$ for all $u \in \mathrm{IN}(F)$.

We have that

$$\mathtt{wt}_i(\kappa) = \#\{u \in \mathrm{in}(F) \mid i \in \kappa(u)\}$$

Theorem 2.2. *Suppose a labeled indexed forest (F,P) is such that there is a $\kappa \in \mathrm{Komp}_P(F)$ with*
$$\mathtt{wt}(\kappa) = \mathfrak{c}(F).$$
Then
$$\widetilde{\mathfrak{P}}_F(x) = \sum_{\kappa \in \mathrm{Komp}_P(F)} \beta^{|\kappa|-|F|} x^{\mathtt{wt}(\kappa)}$$

Definition 2.3. Let $W \in \mathcal{WC}(w)$ be a WC graph for w. Define via the Nadeau-Tewari insertion algorithm
$$(P(W), Q(W)) = (P(\mathrm{word}(W)), Q(\mathrm{word}(W)))$$
and let
$$F(W) = \mathrm{shape}(P(W))$$
Define $\mathfrak{c}(F)$ to be the code of the indexed forest F.

Definition 2.4. Let (F,P) be a labeled indexed forest and let $\kappa \in \mathrm{Komp}_P(F)$. Consider the set
$$A = \bigcup_{v \in \mathrm{IN}(F)} \kappa(v) \times \{P(v)\}$$
Define
$$\mathcal{WC}(\kappa) = \{(a_i, b_i + 1 - a_i) \mid (a_i, b_i) \in A\}$$

Theorem 2.5. *For a permutation w, let $\mathcal{WC}_{\mathfrak{P}}(w,F)$ be the set of $W \in \mathcal{WC}(w)$ such that $\mathfrak{c}_{\mathfrak{P}}(W) = \mathfrak{c}(F)$. Then we have that $\mathcal{WC}(w)$ is the disjoint union of $\mathcal{WC}(\mathrm{Komp}_{P(W)}(F(W)))$ for $W \in \mathcal{WC}_{\mathfrak{P}}(w,F)$.*

Corollary 2.6. *The Grothendieck polynomial $\mathfrak{G}_w^{(\beta)}(x)$ satisfies*
$$\mathfrak{G}_w^{(\beta)}(x) = \sum_{F \in \mathrm{IF}} \beta^{|F|-\ell(w)} a_w^F \widetilde{\mathfrak{P}}_F(x)$$
where
$$a_w^F = \#\{W \in \mathcal{WC}_{\mathfrak{P}}(w,F) \mid \mathtt{wt}(W) = \mathfrak{c}(F)\}$$

Theorem 2.7. *Let a be a composition. Let w be the permutation such that $\mathfrak{c}(w) = a$. Then we have that*
$$\widetilde{\mathfrak{P}}_a(x) = \sum_{W \in \mathcal{WC}_{\mathfrak{P}}(w,a) \cap \mathcal{WC}_{\mathrm{QY}}(w)} \mathcal{G}_{\mathtt{wt}(W)}(x)$$

Definition 2.8. We define an invariant $\widetilde{P}_{\mathfrak{P}}(W)$ of a WC graph W to be the WC graph C such that $\mathfrak{c}_{\mathfrak{P}}(C) = \mathtt{wt}(C) = \mathfrak{c}_{\mathfrak{P}}(W)$ and there exists a $\kappa \in \mathrm{Komp}_{P(C)}(F(C))$ such that $W \in \mathcal{WC}(\kappa)$.

Theorem 2.9 (Dual Grove LR rule)**.** *Suppose*
$$\widetilde{\mathfrak{P}}_a^* \uplus \widetilde{\mathfrak{P}}_b^* = \sum_c P_{ab}^c \widetilde{\mathfrak{P}}_c^*$$
for integers P_{ab}^c, with $\ell(a) = p$ and $\ell(b) = q$. Let C be a WC graph with $\mathtt{ht}(C) = p+q$ such that $\mathfrak{c}_{\mathfrak{P}}(C) = \mathtt{wt}(C) = c$. Then P_{ab}^c is the number of WC graphs W with $\widetilde{P}_{\mathfrak{P}}(W) = \widetilde{P}_{\mathfrak{P}}(C)$ such that $\mathfrak{c}_{\mathfrak{P}}(\mathtt{clip}^p(W)) = \mathtt{wt}(\mathtt{clip}^p(W)) = a$ and $\mathfrak{c}_{\mathfrak{P}}(\mathtt{trim}^p(W)) = \mathtt{wt}(\mathtt{trim}^p(W)) = b$.

3. Hecke insertion, K-Knuth equivalence, and Lascoux classes

The purpose of this section is to attach to every WC graph a *Lascoux class*, namely the K-Knuth equivalence class of the increasing tableau obtained by Hecke insertion of its word. The main input is a theorem, due to Reiner–Yong [5] (conjecture) and Shimozono–Yu [7] (proof), stating that the WC graphs of w lying in a single K-Knuth class have generating function equal to a single Lascoux polynomial. This refines the symmetric statement of Buch–Kresch–Shimozono–Tamvakis–Yong [1]. We phrase everything in the WC-graph language of the previous sections, so that it can be used to show that WC-graph transition, and hence the squash product, preserves Lascoux classes.

Theorem 3.1 ([1, 2])**.** *For words c, c' of positive integers the following hold.*

(1) $\mathsf{P}(c) = \mathsf{P}(c')$ *if and only if* $c \overset{K}{\equiv} c'$*; thus the insertion tableau is a complete invariant of the K-Knuth class.*

(2) *The Hecke product $s_{c_1} \bullet \cdots \bullet s_{c_n}$ depends only on the K-Knuth class of c; it equals the Hecke product of the row reading word of $\mathsf{P}(c)$.*

Let ∂_i denote the ith divided difference operator,

$$\partial_i f = \frac{f - s_i f}{x_i - x_{i+1}},$$

and define the *K-theoretic isobaric (Demazure) operator*

$$\pi_i^{(\beta)} f = \partial_i (x_i (1 + \beta x_{i+1}) f).$$

Then $\pi_i^{(\beta)}(1) = 1$, $\pi_i^{(\beta)}$ sends polynomials to polynomials whose value is symmetric in x_i, x_{i+1}, and at $\beta = 0$ it recovers the Demazure operator $\pi_i f = \partial_i(x_i f)$ of the key polynomials.

Definition 3.2. For a weak composition $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$ the *Lascoux polynomial* $\mathfrak{L}_\alpha$ is defined by:

(1) if $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n$ (i.e. α is a partition), then $\mathfrak{L}_\alpha = x^\alpha = \prod_i x_i^{\alpha_i}$;
(2) if $\alpha_i < \alpha_{i+1}$ for some i, then $\mathfrak{L}_\alpha = \pi_i^{(\beta)} \mathfrak{L}_{s_i \alpha}$, where $s_i \alpha$ interchanges α_i and α_{i+1}.

This is well defined by [7]. The lowest-β component of $\mathfrak{L}_\alpha$ (its terms of total degree $|\alpha|$, i.e. the β^0 part) is the key (Demazure) polynomial κ_α.

3.1. **The Grothendieck-to-Lascoux expansion.** Recall from the definition of a WC graph that the elements of W are listed as $w_1 < w_2 < \cdots < w_n$ in the total order on $\mathbb{N} \times \mathbb{N}$, with $w_i = (a_i, b_i)$, and that

$$s(a_i, b_i) = s_{a_i + b_i - 1}.$$

Definition 3.3. For a WC graph W, define the *diagonal skew tableau* $\mathtt{dw}(W)$ to be the unique tableau of skew shape $\mathrm{st}(|W|)/\mathrm{st}(|W|-1)$ whose reading word is $\mathrm{word}(W)$. The Lascoux invariant $\mathfrak{L}(W)$ of W is the greedy rectification of $\mathtt{dw}(W)$. We can in fact apply this to any word.

Theorem 3.4 (Dual Lascoux LR rule)**.** *Suppose*

$$\mathcal{L}_a^* \uplus \mathcal{L}_b^* = \sum_c L_{ab}^c \mathcal{L}_c^*$$

for integers L_{ab}^c, with $\ell(a) = p$ and $\ell(b) = q$. Let C be a WC graph with $\mathtt{ht}(C) = p + q$ such that $\mathtt{wt}(\mathfrak{L}(C)) = \mathtt{wt}(C) = c$. Then L_{ab}^c is the number of WC graphs W with $\mathfrak{L}(W) = \mathfrak{L}(C)$ such that $\mathtt{wt}(\mathfrak{L}(\mathtt{clip}^p(W))) = \mathtt{wt}(\mathtt{clip}^p(W)) = a$ and $\mathtt{wt}(\mathfrak{L}(\mathtt{trim}^p(W))) = \mathtt{wt}(\mathtt{trim}^p(W)) = b$.

The next theorem is the symmetric statement of Buch–Kresch–Shimozono–Tamvakis–Yong, in the WC-graph normalization: grouping the WC graphs of w by Lascoux class collects their generating function into symmetric Grothendieck building blocks.

Theorem 3.5 ([1])**.** *Fix an increasing tableau P of straight shape λ whose row reading word has Hecke product w. Then*

$$\sum_{\substack{W \in \mathcal{WC}(w) \\ \mathsf{P}(W) = P}} \beta^{|W| - \ell(w)} x^{\mathtt{wt}(W)}$$

is symmetric in the variables it involves; summed over all such P of a fixed shape λ it equals the stable (Grassmannian) Grothendieck contribution $a_{w,\lambda} G_\lambda$, recovering the factor-sequence expansion $G_w = \sum_\lambda a_{w,\lambda} G_\lambda$.

The nonsymmetric refinement replaces each symmetric block by a single Lascoux polynomial. This is the "Grothendieck to Lascoux" expansion.

Theorem 3.6 (Reiner–Yong [5]; Shimozono–Yu [7])**.** *There is a map $P \mapsto \alpha(P)$ assigning to each increasing tableau P a weak composition $\alpha(P)$, with underlying partition* $\mathrm{sh}(P)$*, determined by the K-theoretic right key of P, such that for every increasing tableau P arising as an insertion tableau of some $W \in \mathcal{WC}(w)$,*

$$\sum_{\substack{W \in \mathcal{WC}(w) \\ \mathsf{P}(W) = P}} \beta^{|W| - \ell(w)} x^{\mathtt{wt}(W)} = \mathfrak{L}_{\alpha(P)}.$$

Consequently

$$\mathfrak{G}_w^{(\beta)}(x) = \sum_{P \in \mathcal{I}(w)} \mathfrak{L}_{\alpha(P)} = \sum_{\alpha} c_{w,\alpha}\, \mathfrak{L}_\alpha, \qquad c_{w,\alpha} = \#\{P \in \mathcal{I}(w) \mid \alpha(P) = \alpha\} \in \mathbb{Z}_{\geq 0},$$

where $\mathcal{I}(w) = \{\mathsf{P}(W) \mid W \in \mathcal{WC}(w)\}$ *is the set of Lascoux classes of* w.

The content we use below is the following: *the Lascoux polynomial attached to a WC graph depends only on its Lascoux class.* We isolate this as a corollary.

Corollary 3.7. *The assignment*

$$W \longmapsto \mathfrak{L}_{\alpha(\mathsf{P}(W))}$$

factors through the Lascoux class $[W] = \mathsf{P}(W)$. *In particular, if* Φ *is any map on WC graphs with* $\mathsf{P}(\Phi(W)) \overset{K}{\equiv} \mathsf{P}(W)$ *for all* W, *equivalently* $\mathtt{dw}(\Phi(W)) \overset{K}{\equiv} \mathtt{dw}(W)$, *then*

$$\mathfrak{L}_{\alpha(\mathsf{P}(\Phi(W)))} = \mathfrak{L}_{\alpha(\mathsf{P}(W))}$$

for all W*; that is,* Φ *preserves Lascoux classes and hence the associated Lascoux polynomial.*

Proof. Immediate from Theorem 3.1(1), which identifies $\mathsf{P}(W)$ with the K-Knuth class of $\mathtt{dw}(W)$, and Theorem 3.6, in which α is a function of $\mathsf{P}(W)$ alone. □

Remark. Corollary 3.7 is the K-theoretic, straight-shape analogue of the grove/forest decomposition of Section 3's predecessor: there the WC graphs of w are grouped by the indexed forest $F(W)$ of the Nadeau–Tewari insertion and each fiber contributes $\widetilde{\mathfrak{P}}_{F(W)}$, whereas here they are grouped by the K-Knuth class $\mathsf{P}(W)$ of Hecke insertion and each fiber contributes $\mathfrak{L}_{\alpha(\mathsf{P}(W))}$. Identifying the two groupings, e.g. establishing an equality $\widetilde{\mathfrak{P}}_F = \mathfrak{L}_{\alpha(\mathsf{P})}$ under the correspondence $F \leftrightarrow \mathsf{P}$, would identify grove polynomials with Lascoux polynomials.

4. Bounded WC graphs and the K-theoretic transition

We now equip WC graphs with a row grading and a transition map, mirroring the treatment of bounded RC graphs in the reduced theory of [6]. Everything in this section specializes to those reduced constructions on the subset of RC graphs.

Definition 4.1. A *bounded WC graph* is a WC graph W together with a specified number of rows $n \geq \max\{a : (a,b) \in W \text{ for some } b\}$ subject to $\mathtt{m}(\mathtt{perm}(W)) \leq n$. We write (W, n) for the pair, abuse notation to treat the number of rows as a property of W, and call $\mathtt{ht}(W) = n$ the *height* of W. For $\mathtt{ht}(W) \geq 1$ we define the *trim* $\mathtt{trim}(W)$ to be the bounded WC graph with height $\mathtt{ht}(W) - 1$ and underlying set

$$\mathtt{trim}(W) = \{(a-1, b-1) : (a,b) \in W,\ a \geq 2\},$$

and we set $\uparrow^m W = \{(a+m, b) \mid (a,b) \in W\}$; these are the bounded RC-graph operations of [6] applied to WC graphs verbatim. Let $\mathcal{BWC}$ be the free abelian group on all bounded WC graphs, graded by height,

$$\mathcal{BWC} = \bigoplus_{n=0}^{\infty} \mathcal{BWC}_n.$$

A WC graph with $\ell(\mathtt{perm}(W)) = |W|$ is exactly an RC graph, and on such graphs $\mathtt{trim}$ and $\uparrow^m$ agree with the RC operations. Thus $\mathcal{BRC} \subseteq \mathcal{BWC}$ is a graded subgroup and the projection $\rho : \mathcal{BWC} \to \mathcal{BRC}$ that kills every WC graph with $|W| > \ell(\mathtt{perm}(W))$ is a graded retraction.

4.1. The transition map via marked bumpless pipe dreams.

In the reduced case the map $\mathcal{Z}$ of [6] realizes the Lascoux–Schützenberger transition on RC graphs. Its K-theoretic counterpart is most naturally described through bumpless pipe dreams. Recall the following ingredients.

(1) **Marked bumpless pipe dreams.** Weigandt [8] introduced *marked bumpless pipe dreams* (marked BPDs) and proved that the double Grothendieck polynomial $\mathfrak{G}_w^{(\beta)}$ is the generating function of marked BPDs of w, extending the Lam–Lee–Shimozono formula from K-theory. Let $\mathrm{MBPD}(w)$ denote the set of marked BPDs of w.

(2) **The HSY bijection.** Huang, Shimozono, and Yu [4] construct a weight-preserving bijection between marked BPDs and reverse compatible pairs, the latter being in bijection with not-necessarily-reduced pipe dreams; their bijection generalizes, in the unmarked reduced case, the canonical bijection of Gao and Huang [3]. Reading a bounded WC graph as the compatible sequence of a (not necessarily reduced) pipe dream, this yields a weight-preserving bijection

$$\Psi : \bigcup_n \{W \in \mathcal{BWC}_n : \mathtt{perm}(W) = w\} \longrightarrow \mathrm{MBPD}(w),$$

whose restriction to RC graphs is the Gao–Huang bijection ϕ used in the reduced theory of [6].

(3) **Weigandt transition.** Weigandt's droop/transition operation $\mathbf{t}$ on bumpless pipe dreams removes the bottom row and rearranges the resulting diagram into a valid BPD of one fewer row, realizing the (co)transition recursion for Grothendieck polynomials; it acts on marked BPDs with an empty bottom row [8, 4].

Definition 4.2. Let W be a bounded WC graph with $\mathtt{ht}(W) = n$ whose row n is empty. Define the *K-theoretic transition map*

$$\mathscr{Z}(W) = \Psi^{-1}(\mathbf{t}(\Psi(W))),$$

a bounded WC graph of height $n-1$. We extend $\mathscr{Z}$ to an endomorphism of $\mathcal{BWC}$ by setting $\mathscr{Z}(W) = 0$ when row n is nonempty.

The following properties are the K-theoretic analogues of the reduced-case transition lemma, the trim-commutation lemma, and the zeroing bijection of [6]. We record them as the working hypotheses of the construction; each specializes to its reduced counterpart under ρ, where it is a theorem of [6].

Proposition 4.3. *On RC graphs the map $\mathscr{Z}$ agrees with $\mathcal{Z}$: for every bounded WC graph W with empty last row,*

$$\rho(\mathscr{Z}(W)) = \mathcal{Z}(\rho(W)),$$

and in particular $\mathscr{Z}|_{\mathcal{BRC}} = \mathcal{Z}$.

Proof sketch. On RC graphs Ψ is the Gao–Huang bijection ϕ and $\mathbf{t}$ restricts to Weigandt's transition on reduced BPDs, so $\mathscr{Z} = \phi^{-1} \circ \mathbf{t} \circ \phi$ there. The reduced-case identity $\mathcal{Z} \circ \phi^{-1} = \phi^{-1} \circ \mathbf{t}$ (the conjecture of [6] relating $\mathcal{Z}$ and $\mathbf{t}$ through ϕ) gives the claim; ρ discards the non-reduced terms, none of which are produced from a reduced input. $\square$

Proposition 4.4. *Let W be a bounded WC graph with $\mathtt{ht}(W) = n \geq 1$ and empty row n. Then $\mathtt{ht}(\mathscr{Z}(W)) = n-1$, the weight is preserved,*

$$\mathtt{wt}(\mathscr{Z}(W)) = \mathtt{wt}(W),$$

and the permutation undergoes the K-theoretic (Grothendieck) transition: $\mathtt{perm}(\mathscr{Z}(W))$ is obtained from $\mathtt{perm}(W)$ by a single step of the Grothendieck cotransition recursion in row n.

Proposition 4.5. *Let (W, n) be a bounded WC graph with $n \geq 2$ whose row n is empty. Then*

$$\mathscr{Z}(\mathtt{trim}(W)) = \mathtt{trim}(\mathscr{Z}(W)).$$

Proof sketch. This is the K-analogue of the trim-commutation lemma of [6]. Trimming preserves a suffix of the word of W, and both Ψ and $\mathbf{t}$ act on the bottom rows of the marked BPD without reference to the top row: the transition $\mathbf{t}$ proceeds downward from the removed bottom row, so discarding the top row commutes with it. Under ρ the identity specializes to the reduced statement of [6]. $\square$

Proposition 4.6. *Let w be a permutation with last descent at most n, and let $\mathcal{WC}(w,n)^0$ be the set of bounded WC graphs W with $\mathtt{ht}(W) = n$, $\mathtt{perm}(W) = w$, and empty row n. Then*

$$\mathscr{Z} : \mathcal{WC}(w,n)^0 \longrightarrow \bigcup_{w \overset{n}{\searrow} w'} \mathcal{WC}(w', n-1)$$

is a weight-preserving bijection, where w' ranges over the outputs of the Grothendieck cotransition in row n. Consequently

$$\mathscr{Z}\Big(\sum_{\substack{\mathtt{perm}(W)=w \\ \mathtt{ht}(W)=n}} W\Big) = \sum_{w \overset{n}{\searrow} w'} \sum_{\substack{\mathtt{perm}(W')=w' \\ \mathtt{ht}(W')=n-1}} W',$$

the WC-graph form of the Grothendieck transition formula.

5. WC Graph ring

We now lift the K-theoretic transition map $\mathscr{X}$ to an associative product on $\mathcal{BWC}$, in exact parallel with the product $\uplus$ on $\mathcal{BRC}$ built from $\mathcal{Z}$ in [6]. Only the underlying zeroing map has changed: $\mathtt{trim}^p$ is unchanged, and $\mathtt{clip}^p$ is now built from $\mathscr{X}$.

Definition 5.1. Let W be a bounded WC graph with $\mathtt{ht}(W) = n$ and let $0 \le p \le n$. Define $\mathtt{trim}^p(W) \in \mathcal{BWC}_{n-p}$ to be the result of discarding the top p rows of W and reindexing (equivalently, the p-fold iterate of $\mathtt{trim}$), and define $\mathtt{clip}^p(W) \in \mathcal{BWC}_p$ to be the $(n-p)$-fold iterate of $\mathscr{X}$ applied to the bottom row,

$$\mathtt{clip}^p(W) = \mathscr{X}^{\,n-p}(W),$$

which is defined whenever each intermediate bottom row is empty at the moment it is zeroed (and is 0 otherwise). Thus $\mathtt{clip}^p(W)$ and $\mathtt{trim}^p(W)$ decompose W across the horizontal cut between rows p and $p+1$.

Associativity of the product below rests on two structural identities relating $\mathtt{clip}^p$ and $\mathtt{trim}^p$: functoriality of $\mathtt{clip}$ under iteration, and commutativity of $\mathtt{clip}$ with $\mathtt{trim}$. The first is formal; the second is the content of Proposition 4.5.

Lemma 5.2. *Let W be a bounded WC graph with $\mathtt{ht}(W) = n$ and $0 \le p \le p+q \le n$. Then*

$$\mathtt{clip}^p(\mathtt{clip}^{p+q}(W)) = \mathtt{clip}^p(W).$$

Proof. Both sides are computed by iterating $\mathscr{X}$ on the bottom row. With $r = n-(p+q)$, the graph $\mathtt{clip}^{p+q}(W)$ is the r-fold iterate $\mathscr{X}^r(W)$ and $\mathtt{clip}^p(W) = \mathscr{X}^{r+q}(W)$; the first r iterations in the computation of $\mathtt{clip}^p(W)$ produce $\mathtt{clip}^{p+q}(W)$, and the remaining q produce $\mathtt{clip}^p(\mathtt{clip}^{p+q}(W))$. Hence the two agree. □

Lemma 5.3 (Commutativity of $\mathtt{clip}$ and $\mathtt{trim}$)**.** *Let W be a bounded WC graph with $\mathtt{ht}(W) = p+q+r$. Then*

$$\mathtt{trim}^p(\mathtt{clip}^{p+q}(W)) = \mathtt{clip}^q(\mathtt{trim}^p(W)).$$

Proof. Both sides have height q. By Proposition 4.5, $\mathscr{X}(\mathtt{trim}(W')) = \mathtt{trim}(\mathscr{X}(W'))$ whenever the last row of W' is empty. Since $\mathtt{clip}^{p+q}$ on a graph of height $p+q+r$ and $\mathtt{clip}^q$ on a graph of height $q+r$ are both the r-fold iterate of $\mathscr{X}$ on the bottom row (cf. Lemma 5.2), iterating Proposition 4.5 r times gives

$$\mathtt{trim}^p(\mathtt{clip}^{p+q}(W)) = \mathtt{trim}^p(\mathscr{X}^r(W)) = \mathscr{X}^r(\mathtt{trim}^p(W)) = \mathtt{clip}^q(\mathtt{trim}^p(W)),$$

as required. □

Definition 5.4. For bounded WC graphs W_1 with $\mathtt{ht}(W_1) = m$ and W_2 with $\mathtt{ht}(W_2) = n$, define

$$W_1 \uplus W_2 = \sum_{\substack{\mathtt{clip}^m(W')=W_1 \\ \mathtt{trim}^m(W')=W_2}} W',$$

where W' ranges over bounded WC graphs with $\mathtt{ht}(W') = m+n$, and extend bilinearly to $\mathcal{BWC}$.

Theorem 5.5. *The product $\uplus$ is associative, and $\mathcal{BWC}$ is a ring under $\uplus$ with identity the empty WC graph of height* 0*.*

Proof. Let W_1, W_2, W_3 be bounded WC graphs of heights p, q, r, and let W be a bounded WC graph with $\mathtt{ht}(W) = p+q+r$. Unwinding Definition 5.4 in each bracketing, W contributes to the doubly-bracketed products under the conditions

	Left $(W_1 \uplus W_2) \uplus W_3$	Right $W_1 \uplus (W_2 \uplus W_3)$
(i)	$\mathtt{clip}^p(\mathtt{clip}^{p+q}(W)) = W_1$	$\mathtt{clip}^p(W) = W_1$
(ii)	$\mathtt{trim}^p(\mathtt{clip}^{p+q}(W)) = W_2$	$\mathtt{clip}^q(\mathtt{trim}^p(W)) = W_2$
(iii)	$\mathtt{trim}^{p+q}(W) = W_3$	$\mathtt{trim}^q(\mathtt{trim}^p(W)) = W_3$

Row (i) is equivalent on both sides by Lemma 5.2, row (ii) by Lemma 5.3, and row (iii) because $\mathtt{trim}^{p+q} = \mathtt{trim}^q \circ \mathtt{trim}^p$ by definition of $\mathtt{trim}$. Each condition is an equality carrying no combinatorial multiplicity, so W occurs with multiplicity 1 in either bracketing, and the two bracketed products agree. Distributivity is imposed by bilinear extension, and the empty height-0 WC graph is a two-sided identity. $\square$

Because $\mathscr{X}$ restricts to $\mathcal{Z}$ on RC graphs (Proposition 4.3) and $\mathtt{trim}$ is unchanged, the retraction ρ intertwines the two clip/trim decompositions; hence it is compatible with the products.

Proposition 5.6. *The linear map*

$$\rho : \mathcal{BWC} \to \mathcal{BRC}$$

such that

$$\rho(W) = \begin{cases} W & \text{if } W \text{ is an RC graph} \\ 0 & \text{otherwise} \end{cases}$$

is a surjective ring homomorphism.

6. The K-plactic monoid and algebra

The reduced theory of [6] attaches to each rank k a *plactic monoid* Plac_k: the set of height-k RC graphs whose permutation is k-Grassmannian, made into a monoid by the squash product, which is a combinatorial model for the ring of Schur polynomials in k variables. We now introduce its K-theoretic, set-valued analogue. It is obtained by replacing RC graphs with WC graphs and, correspondingly, semistandard Young tableaux and Schur polynomials with their set-valued and stable Grothendieck counterparts.

Recall that a permutation w is k-*Grassmannian* if it is either the identity or has a unique descent, located at position k. For such a w the associated partition

$$\lambda(w) = (w(k) - k,\ w(k-1) - (k-1),\ \ldots,\ w(1) - 1)$$

has at most k parts, and the Grothendieck polynomial $\mathfrak{G}_w^{(\beta)}$, expressed in the variables $x_1, \ldots, x_k$, is the stable (Grassmannian) Grothendieck polynomial $G_{\lambda(w)}(x_1, \ldots, x_k)$; its lowest-degree component is the Schur polynomial $s_{\lambda(w)}(x_1, \ldots, x_k)$. By the WC-graph formula $\mathfrak{G}_w^{(\beta)}(x) = \sum_{W \in \mathcal{WC}(w)} \beta^{|W| - \ell(w)} x^{\mathtt{wt}(W)}$, the height-$k$ WC graphs of a k-Grassmannian permutation are the set-valued analogue of the semistandard tableaux enumerated by a Schur polynomial.

Definition 6.1 (Squash product)**.** Let W_1 and W_2 be bounded WC graphs of height k. Their *disjoint union* $W_1 \star W_2$ is formed by shifting every entry of W_2 upward far enough that each reflection $s(a, b)$ used by W_2 exceeds—and hence Hecke commutes with—every reflection used by W_1, and then overlaying the two graphs row by row. The weight is additive, $\mathtt{wt}(W_1 \star W_2) = \mathtt{wt}(W_1) + \mathtt{wt}(W_2)$. The *squash product* is

$$W_1 \boxtimes W_2 = \mathtt{clip}^k(W_1 \star W_2),$$

the clip of the disjoint union back down to k rows (Definition 5.1).

On k-Grassmannian factors the squash product is exactly the single surviving term of the ring product $\uplus$ of Definition 5.4: since the clip $\mathtt{clip}^k$ and trim $\mathtt{trim}^k$ of $W_1 \star W_2$ recover W_1 and W_2 respectively,

$$W_1 \boxtimes W_2 = W_1 \uplus W_2 \qquad (\mathtt{ht}(W_1) = \mathtt{ht}(W_2) = k).$$

In particular $\boxtimes$ is associative, being the restriction of the associative product $\uplus$ (Theorem 5.5) to n-Grassmannian WC graphs.

Definition 6.2 (K-plactic monoid and algebra)**.** The *K-plactic monoid of rank k* is the set

$$\mathrm{Plac}_k^K = \{\, W \in \mathcal{BWC}_k : \mathtt{perm}(W) \text{ is } k\text{-Grassmannian} \,\}$$

of height-k WC graphs whose permutation has no descent or a single descent at k, equipped with the squash product $\boxtimes$. It is a monoid with unit the empty height-k WC graph. The *K-plactic algebra* $\mathbb{Z}\mathrm{Plac}_k^K$ is its monoid algebra: the free abelian group on Plac_k^K with the bilinear extension of $\boxtimes$.

Theorem 6.3. *The set* Plac_k^K *is closed under* $\boxtimes$*, and* $(\mathrm{Plac}_k^K, \boxtimes)$ *is an associative monoid. It is the set-valued analogue of the plactic monoid* Plac_k *of semistandard tableaux of at most k rows: the degeneration that discards every WC graph W with $|W| > \ell(\mathtt{perm}(W))$ (equivalently, restricts to RC graphs) sends* Plac_k^K *onto* Plac_k *and* $\boxtimes$ *to the classical squash product.*

Proof. Closure holds because $\texttt{clip}^k$ carries a height-$2k$ WC graph whose top and bottom halves are k-Grassmannian to a height-k WC graph whose permutation is again k-Grassmannian: the transition $\mathscr{X}$ (Proposition 4.4) preserves the property of having a single descent at the top row. Associativity is the restriction of Theorem 5.5. The degeneration is the restriction of the retraction $\rho : \mathcal{BWC} \to \mathcal{BRC}$ (which is a ring homomorphism), and on RC graphs $\texttt{clip}^k = \mathcal{Z}^k$ agrees with the reduced clip of [6], so ρ intertwines $\boxtimes$ with the reduced squash product. $\square$

The squash product realizes stable Grothendieck multiplication, in exact parallel with the crystal isomorphism of the reduced theory [6]. We record this as the K-theoretic analogue.

Theorem 6.4. *Let u, v be k-Grassmannian permutations. Then the disjoint union induces a weight-preserving bijection*

$$\{W \in \mathrm{Plac}_k^K : \texttt{perm}(W) = u\} \times \{W' \in \mathrm{Plac}_k^K : \texttt{perm}(W') = v\} \;\longrightarrow\; \{U \in \mathcal{BWC}_{2k} : \texttt{clip}^k(U) \in \mathrm{Plac}_k^K(u),\ \texttt{trim}^k(U) \in \mathrm{Plac}_k^K($$

and clipping back to k rows decomposes the result according to the K-theoretic Littlewood–Richardson (Buch) rule [1]*: writing λ, μ for the partitions of u, v,*

$$\sum_{\substack{W \in \mathrm{Plac}_k^K(u) \\ W' \in \mathrm{Plac}_k^K(v)}} W \boxtimes W' = \sum_{\nu} c_{\lambda\mu}^{\nu}(\beta) \sum_{\substack{W'' \in \mathrm{Plac}_k^K \\ \lambda(\texttt{perm}(W''))=\nu}} W'',$$

where $c_{\lambda\mu}^{\nu}(\beta)$ are the structure constants of the stable Grothendieck polynomials, $G_\lambda G_\mu = \sum_\nu c_{\lambda\mu}^{\nu}(\beta) G_\nu$.

For a k-Grassmannian permutation w we assemble the WC graphs of w into a single element of the K-plactic algebra.

Definition 6.5. For a k-Grassmannian permutation w, the *Grothendieck element* is

$$\mathbf{G}_w = \sum_{\substack{W \in \mathrm{Plac}_k^K \\ \texttt{perm}(W)=w}} W \;\in\; \mathbb{Z}\mathrm{Plac}_k^K,$$

the sum of all height-k WC graphs with permutation w. Under the weight map $W \mapsto \beta^{|W|-\ell(w)} x^{\texttt{wt}(W)}$ it maps to the stable Grothendieck polynomial $G_{\lambda(w)}(x_1, \ldots, x_k)$.

Theorem 6.6. *As w ranges over the k-Grassmannian permutations, the Grothendieck elements $\mathbf{G}_w$ are linearly independent and span a commutative subring $\mathbf{G}_k \subseteq \mathbb{Z}\mathrm{Plac}_k^K$. The weight map induces a ring isomorphism*

$$\mathbf{G}_k \xrightarrow{\;\sim\;} \mathbb{Z}[G_\lambda(x_1, \ldots, x_k) : \lambda \textit{ a partition with} \le k \textit{ parts}], \qquad \mathbf{G}_w \longmapsto G_{\lambda(w)}(x_1, \ldots, x_k),$$

onto the ring generated by the stable Grothendieck polynomials in k variables. Passing to lowest degree recovers the classical model: $\mathbf{G}_k$ degenerates to the ring of symmetric polynomials spanned by the Schur polynomials $s_\lambda(x_1, \ldots, x_k)$, and $\mathbf{G}_w$ to the Schur element of Plac_k.

Proof. Distinct k-Grassmannian permutations have disjoint sets of WC graphs, so the $\mathbf{G}_w$ are linearly independent. By Theorem 6.4 the squash product of two Grothendieck elements decomposes with the same structure constants as the product of the corresponding stable Grothendieck polynomials,

$$\mathbf{G}_u \boxtimes \mathbf{G}_v = \sum_w c_{u,v}^{w}(\beta)\, \mathbf{G}_w, \qquad G_{\lambda(u)}\, G_{\lambda(v)} = \sum_w c_{u,v}^{w}(\beta)\, G_{\lambda(w)},$$

where $c_{u,v}^{w}(\beta)$ are the K-theoretic Littlewood–Richardson (Buch) coefficients. Hence the span $\mathbf{G}_k$ is a subring, the assignment $\mathbf{G}_w \mapsto G_{\lambda(w)}$ is a ring homomorphism, and it is bijective onto the stated ring because the stable Grothendieck polynomials of partitions with at most k parts are a $\mathbb{Z}$-basis of it. Commutativity follows since the target ring is commutative. Setting $\beta = 0$ (equivalently, retaining the lowest-degree part, which is ρ followed by symmetrization) sends $G_\lambda \mapsto s_\lambda$ and recovers the plactic model of [6]. $\square$

Remark. Theorem 6.6 is the K-theoretic analogue of the statement that, inside the plactic algebra, the sums of tableaux of fixed shape span a commutative ring isomorphic to the ring of Schur polynomials. Here shapes are replaced by k-Grassmannian permutations, tableaux by WC graphs, Knuth equivalence by K-Knuth equivalence (Section 3), and Schur polynomials by stable Grothendieck polynomials.

The K-plactic algebras also carry the degeneration to the reduced plactic algebras and assemble into a tensor model for $\mathcal{BWC}_n$.

Definition 6.7. For each $n > 0$ there is a surjective ring homomorphism

$$\mathbb{Z}\mathrm{Plac}_n^K \to \mathbb{Z}\mathrm{Plac}_n$$

obtained on elementary symmetric WC graphs by sending $E^{\beta}_{p,k}$ with $|\beta|> p$ to $E^{\beta}_{|\beta|,k}$; it is the restriction of the retraction ρ and provides a section of the inclusion $\mathrm{Plac}_n \hookrightarrow \mathrm{Plac}_n^K$. Define

$$\mathcal{M}_n^K = \bigotimes_{i=1}^{n} \mathbb{Z}\mathrm{Plac}_i^K.$$

There is an evident homomorphism $\mathcal{M}_n^K \to \mathcal{M}_n$ obtained by applying the above homomorphism to each factor, giving a commutative diagram

$$\begin{tikzcd} \mathcal{M}_n^K \arrow[r] \arrow[d] & \mathcal{M}_n \arrow[d] \\ \mathcal{BWC}_n \arrow[r, dashed] & \mathcal{BRC}_n \end{tikzcd}$$

Remark. The compatibility of $\uplus$ with the Lascoux decomposition of Theorem 3.6 is deferred. Corollary 3.7 shows that any map preserving the K-Knuth class $\mathsf{P}(W)$ preserves the associated Lascoux polynomial; the outstanding question is how $\mathscr{X}$, and hence $\mathtt{clip}^p$ and the squash product, interact with the classes $\mathsf{P}(W)$. We return to this once the properties of $\mathscr{X}$ in Propositions 4.4–4.6 are established.

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7. Formulas

Define the double grothendieck polynomial $\mathfrak{G}_w(x; y)$ by

$$\mathfrak{G}_w^{(\beta)}(x; y) = \sum_{W\in\mathcal{WC}(w)} \beta^{|W|-\ell(w)}(x \ominus y)^W$$

where

$$(x \ominus y)^W = \prod_{(i,j)\in W} \left(\frac{x_i - y_j}{1 + \beta y_j}\right)$$

$$\omega^i = \frac{(1 + \beta t_i) - s_i(1 + \beta t_i)}{t_i - t_{i+1}} = \frac{(1 - s_i)(1 + \beta t_i)}{t_i - t_{i+1}} = \delta^i(1 + \beta t_i)$$

where δ^i is the divided difference operator on the t variables. Leibniz rule,

$$\omega^i(fg) = \omega^i(f)g + f(\delta^i + s_i)g$$

Let $\mathbf{w}_1 \cdots \mathbf{w}_n$ be a reduced word for w. Let $\mathbf{u}_1 \cdots \mathbf{u}_n$ be a subword, meaning $\mathbf{u}_i \in \{1, \mathbf{w}_i\}$. Let $\mathbf{v}$ be a second subword in the complementary spaces of $\mathbf{u}$, meaning $\mathbf{v}_i \in \{1, \mathbf{w}_i\}$ and $\mathbf{v}_i = 1$ if $\mathbf{u}_i = \mathbf{w}_i$. Then let

$$\omega^{\mathbf{u}} = \omega^{\mathbf{u}_1} \cdots \omega^{\mathbf{u}_n}$$

and let

$$\pi^i = \begin{cases} \omega^i & \text{if } \mathbf{v}_i = \mathbf{w}_i \\ 1 & \text{if } \mathbf{u}_i = \mathbf{w}_i \\ \delta^i + s_i & \text{if } \mathbf{u}_i = 1 \text{ and } \mathbf{v}_i = 1 \end{cases}$$

and let

$$\omega_{\mathbf{v}}^{\mathbf{w}/\mathbf{u}} = \pi^1 \cdots \pi^n$$

Then we have the following general Leibniz rule:

$$\omega^w(fg) = \sum_{\mathbf{u},\mathbf{v}} \omega^{\mathbf{u}}(f)\omega_{\mathbf{v}}^{\mathbf{w}/\mathbf{u}}(g)$$

where the sum is over all subwords $\mathbf{u}$ and $\mathbf{v}$ in complementary spaces of $\mathbf{u}$.

To make this independent of subwords, we can define ω_u^w as

$$\omega_u^w = \sum_{\mathbf{u},\mathbf{v}} \beta^{|\mathbf{u}|-\ell(u)} \omega_{\mathbf{v}}^{\mathbf{w}/\mathbf{u}}$$

where the sum is over all subwords $\mathbf{u}$ of w whose Hecke product is u, with $\mathbf{v}$ in the complementary spaces of $\mathbf{u}$. Then we have

$$\omega^w(fg) = \sum_u \omega^u(f)\omega_u^w(g)$$

$$\partial^i \frac{y_i}{1+\beta y_i} = \frac{\frac{y_i}{1+\beta y_i} - \frac{y_{i+1}}{1+\beta y_{i+1}}}{y_i - y_{i+1}}$$

$$= \frac{\frac{y_i}{(}1+\beta y_{i+1}) - y_{i+1}(1+\beta y_i)(1+\beta y_i)(1+\beta y_{i+1})}{y_i - y_{i+1}}$$

$$= \frac{1}{(1+\beta y_i)(1+\beta y_{i+1})}$$

$$+s_i = \frac{1+y_{i+1}}{(1+\beta y_i)(1+\beta y_{i+1})}$$

$$= \frac{1+\beta y_{i+1} - (\beta-1)y_{i+1}}{(1+\beta y_i)(1+\beta y_{i+1})}$$

$$= \frac{1}{1+\beta y_i} - \frac{(\beta-1)y_{i+1}}{(1+\beta y_i)(1+\beta y_{i+1})}$$

$\partial^i \otimes 1 + s_i \otimes \partial^i + s_i \otimes s_i \; \partial^i \otimes 1 + s_i \otimes (\partial^i + s_i)$
$(\partial^i + s_i) \otimes 1 + s_i \otimes (\partial^i + s_i) - s_i \otimes 1$

$$\pi^i = \frac{(1+x_{i+1}) - (1+x_i)s_i}{x_i - x_{i+1}}$$

$$\partial^i(1+x_{i+1}) = (1+y_1)\partial^i(x_{i+1} \ominus y_1) + y_1\partial^i = (x_i \ominus y_1)\partial^i + y_1\partial^i - 1$$

$$\mathfrak{G}_\mu(x;y) = y^{-|W|}\,\mathfrak{S}_\mu(x;y)$$

$$\pi^p \cdots \pi^1 = \partial^p \cdots \partial^1 E_p(x;1)$$

Write

$$w = c[\mathfrak{c}_n^*(w);n] \cdots c[\mathfrak{c}_1^*(w);1]$$

$$\mathfrak{S}_\mu(y;1)^{-1}\partial^{\mathfrak{c}_1^*(w)} \sum E_{p-\ell(\mu,v)}(y_{P(\mu,v)};1)\,\mathfrak{S}_v(x;y)$$

$$\pi^i = \frac{1}{x_i - x_{i+1}}((1+x_{i+1}) - (1+x_i)s_i)$$

$$(1+\beta x_i)\partial^i - \beta$$

$$\mathfrak{G}_\mu(x;y) = (1 \ominus 0)^W\,\mathfrak{S}_\mu(x;y)$$

$$\mathfrak{G}_\mu(x;y) = (1 \ominus 0)^W\,\mathfrak{S}_\mu(x;y)$$
$$\partial^i(1 + x_{i+1} - x_i + x_i)$$

$$(1+y)^\mu \partial^i (1+\beta x_{i+1})\,\mathfrak{S}_\mu(x;y)$$

Claim: Every Grothendieck polynomial $\mathfrak{G}_w(x)$ can be expressed as a sum of products of ordinary elementary symmetric stable Grothendieck polynomials with integer coefficients, with the number of factors of the product fixed (allowing degree 0 factors) each being in a fixed number of x variables.

Let $\mathfrak{K}^i$ denote the operator $(1+y_{i+1})\partial^i - 1$, acting on the y variables. Consider the product of operators

$$\mathfrak{K}^{i+p} \cdots \mathfrak{K}^{i+1}\mathfrak{K}^i = ((1+y_{i+p})\partial^{i+p} - 1) \cdots ((1+y_{i+1})\partial^{i+1} - 1)((1+y_i)\partial^i - 1)E_p(\tau^i y;1)$$

We have that the top degree factorial elementary stable Grothendieck polynomial in k variables is

$$\mathfrak{E}_k(x;y) = E_k\left(\frac{x}{1+x};y_1\right)E_k(x;1)$$

where E_k is the ordinary top degree factorial elementary symmetric polynomial. Thus, for a dominant permutation, applying such a product of operators gives

$$\mathfrak{K}^{i+p}\cdots\mathfrak{K}^{i+1}\mathfrak{K}^i\prod_{j=1}\mathfrak{E}_{\lambda_j}(x;y_j)$$

$$= \partial^{i+p}\cdots\partial^{i+1}\partial^i E_p(\tau^i y;1)\prod_{j=1}E_{\lambda_j}\left(\frac{x}{1+x};y_j\right)E_{\lambda_j}(x;1)$$

Applying the Schubert formulas and setting $y_i = 0$ yields a sum of products of ordinary elementary symmetric stable Grothendieck polynomials with the same length as λ with the ith factor having λ_i x variables.

Abbreviate such a strip by $\mathfrak{K}^{[i;p]}$. For each permutation u, we have that

$$\mathfrak{K}^u = \mathfrak{K}^{[1;\mathfrak{c}_1(u)]}\mathfrak{K}^{[2;\mathfrak{c}_2(u)]}\cdots\mathfrak{K}^{[n;\mathfrak{c}_n(u)]}$$

Iterating the above argument, we obtain the claim.

$X_i = X_i - 1$

$Y_i = y_i + 1$

$$x_i + y_j + x_iy_j$$

$$h_1^2 - h_2 + h_1$$

$$x_i \oplus y_j = Y_j(X_i - \frac{1}{Y_j})$$

$$Z_j^{-1}(X_i - Z_j)$$

$$\sum_i (1+y_1)^i E_{i,k}(x;y)E_{k-i}(y(1+y_1);-y_1)$$

$$\prod_{i=1}^{n-1}E_{n-i}(x\oplus y_i)$$

$$\mathfrak{G}_{w_0}(x;y) = \sum_{u\in S_n}\partial^{w_0w^{-1}}_{w_0u^{-1}}(\mathfrak{G}_{w_0}(x;y))\mid_{x=y}\mathfrak{G}_u(x;y)$$

8. Abstract Pieri rings

The w_0 for them

$$\widetilde{\mathfrak{G}}_{w_0}(x;t) = \prod_{(i,j)\in W}\frac{x_i - t_j}{1+\beta t_j}$$

Us

$$\mathfrak{G}_{w_0}(x;y) = \beta^{-\ell(w_0)}\prod_{(i,j)\in W}\beta(x_i + y_j + \beta x_iy_j)$$

$$t_j = \frac{-y_j}{1+\beta y_j}$$

$$\mathfrak{G}_{w_0}(x;y) = \beta^{-\ell(w_0)}\prod_{(i,j)\in W}\beta(x_i - \frac{t_j}{1+\beta t_j} - \beta\frac{x_it_j}{1+\beta t_j})$$

$$\begin{aligned}\mathfrak{G}_{(1,3,2)}(x;y) &= y_1^2y_2^2\beta^3 + 2y_1^2y_2\beta^2 + y_1^2\beta + 2y_1y_2^2\beta^2 + 4y_1y_2\beta + 2y_1 + y_2^2\beta \\ &\quad + 2y_2 + \left(y_1y_2\beta^3 + y_1\beta^2 + y_2\beta^2 + \beta\right)\mathfrak{S}_{(2,\ 3,\ 1)}(x;y) \\ &\quad + \left(y_1^2y_2\beta^3 + y_1^2\beta^2 + 2y_1y_2\beta^2 + 2y_1\beta + y_2\beta + 1\right)\mathfrak{S}_{(1,\ 3,\ 2)}(x;y)\end{aligned}$$

$$\mathfrak{G}_{(2,1)}(x;y) = y_1^2\beta + 2y_1 + (y_1\beta + 1)\,\mathfrak{S}_{(2,\ 1)}(x;y)$$

$$\begin{aligned}\mathfrak{G}_{(2,3,1)}(x;y) &= y_1^3y_2\beta^2 + y_1^3\beta + 3y_1^2y_2\beta + 2y_1^2 + 2y_1y_2 \\ &\quad + \left(y_1^2\beta^2 + 2y_1\beta + 1\right)\mathfrak{S}_{(2,\ 3,\ 1)}(x;y) + \left(y_1^3\beta^2 + 3y_1^2\beta + 2y_1\right)\mathfrak{S}_{(1,\ 3,\ 2)}(x;y)\end{aligned}$$

$$\begin{aligned}\mathfrak{G}_{(3,1,2)}(x;y) &= y_1^3y_2\beta^2 + y_1^3\beta + 3y_1^2y_2\beta + 2y_1^2 + 2y_1y_2 + \left(y_1y_2\beta^2 + y_1\beta + y_2\beta + 1\right)\mathfrak{S}_{(3,\ 1,\ 2)}(x;y) \\ &\quad + \left(y_1^2y_2\beta^2 + y_1^2\beta + y_1y_2^2\beta^2 + 4y_1y_2\beta + 2y_1 + y_2^2\beta + 2y_2\right)\mathfrak{S}_{(2,\ 1)}(x;y)\end{aligned}$$

$$\begin{aligned}\mathfrak{G}_{(3,2,1)}(x;y) &= y_1^4y_2^2\beta^3 + 2y_1^4y_2\beta^2 + y_1^4\beta + 4y_1^3y_2^2\beta^2 + 6y_1^3y_2\beta + 2y_1^3 + 5y_1^2y_2^2\beta + 4y_1^2y_2 \\ &\quad + 2y_1y_2^2 + \left(y_1^2y_2\beta^3 + y_1^2\beta^2 + 2y_1y_2\beta^2 + 2y_1\beta + y_2\beta + 1\right)\mathfrak{S}_{(3,\ 2,\ 1)}(x;y) \\ &\quad + \left(y_1^3y_2\beta^3 + y_1^3\beta^2 + 3y_1^2y_2\beta^2 + 3y_1^2\beta + 2y_1y_2\beta + 2y_1\right)\mathfrak{S}_{(3,\ 1,\ 2)}(x;y) \\ &\quad + \left(y_1^4y_2\beta^3 + y_1^4\beta^2 + 4y_1^3y_2\beta^2 + 3y_1^3\beta + 5y_1^2y_2\beta + 2y_1^2 + 2y_1y_2\right)\mathfrak{S}_{(1,\ 3,\ 2)}(x;y) \\ &\quad + \left(y_1^3y_2^2\beta^3 + 2y_1^3y_2\beta^2 + y_1^3\beta + 3y_1^2y_2^2\beta^2 + 5y_1^2y_2\beta + 2y_1^2 + 2y_1y_2^2\beta + 2y_1y_2\right)\mathfrak{S}_{(2,\ 1)}(x;y) + \left(y_1^3y_2\beta^3\right. \\ &\quad \left. + y_1^3\beta^2 + y_1^2y_2^2\beta^3 + 5y_1^2y_2\beta^2 + 3y_1^2\beta + 2y_1y_2^2\beta^2 + 6y_1y_2\beta + 2y_1 + y_2^2\beta + 2y_2\right)\mathfrak{S}_{(2,\ 3,\ 1)}(x;y)\end{aligned}$$

[6]

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